BLUE TOKAI

COFFEE SHOP SALES ANALYSIS

Power BI Dashboard & Business Insights

PROJECT REPORT

This report documents the data model, DAX measures, dashboard design, analytical questions and business observations developed for the Blue Tokai Coffee Shop Sales Analysis project.

Project detail	Description
Tool	Microsoft Power BI Desktop
Analysis period	January–June 2023
Data	Transaction-level sales data for three store locations
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Report date	9 November 2026
Deliverable	Interactive Power BI dashboard and written analysis

1. Executive Summary

This project analyses coffee shop transaction data for January to June 2023. The Power BI dashboard brings together headline KPIs, daily and monthly sales trends, category and product performance, weekday-versus-weekend sales, and order contribution by store.

Overall result: total sales of 698.81K from approximately 149K orders and 214K items. Average order value is 4.69 and average items per order is 1.44.

The main business story: sales grew across the six months, orders are spread almost evenly across the three stores, and a small group of drinks produces most of the revenue. With only 1.44 items per order, the clearest opportunity is to grow basket size through relevant add-ons and bundles rather than relying only on more customers.

Project objectives

- Summarise overall sales, order volume, quantity sold and average customer spend.
- Understand how sales change over time and compare each month with the preceding month.
- Compare sales between weekdays and weekends.
- Evaluate order contribution across store locations.
- Identify leading product categories and product types, and turn the findings into practical recommendations.

Scope and interpretation

The analysis covers January through June 2023. Values shown with K are rounded dashboard displays, so figures derived from them (for example, shares of total) are approximate. Exact values are available in the dashboard tooltips.

2. Data and Model Preparation

The analysis uses a transaction-level table named **Transactions** and a separate calendar table named **Date**. A dedicated date table supports consistent time-based analysis and month filtering.

Transaction fields used in the model

Field	Role in the analysis
transaction_id	Identifies individual transactions; supports order counting.
transaction_date	Transaction date; connects each record to the Date table.
transaction_time	Time of day; available for future time-of-day analysis.
transaction_qty	Quantity purchased; supports total quantity and basket-size measures.
unit_price	Unit selling price; used in sales calculations.
product_id	Product identifier.
product_category	Product category; used for the category ranking visual.
product_type	Product type; used for the Top 10 product ranking.
product_detail	More detailed product description.
store_id	Store identifier.
store_location	Store name (Hell's Kitchen, Astoria, Lower Manhattan); used for store comparison.
Sales	Sales value used by the Total Sales measure.

Date table and relationship

- A dedicated Date table provides a continuous calendar for time intelligence.
- The Date table includes the Date field and calculated attributes such as **Day Type** (Weekday/Weekend) and **Month**.
- A month-start calculated column groups dates at month level for the trend axes.
- The relationship runs from Date[Date] (one side) to Transactions[transaction_date] (many side), single direction and active.
- Both relationship columns use the Date data type, and the Date table is marked as a date table.

This model allows the Month slicer and time-intelligence measures to filter transaction records through the calendar table.

3. DAX Measures and KPI Design

Measures calculate business results dynamically in the current filter context, so the Month slicer changes the KPI values and the visuals that respond to it.

Measure / KPI	Purpose
Total Sales	Total revenue for the selected period.
Total Orders	Number of orders (transactions).
Total Quantity	Total items sold.
Average Order Value	Average sales value per order (4.69).
Average Items per Order	Average quantity per order (1.44); indicates basket size.
Previous Month Sales	Sales for the preceding month, using the Date table.
Previous Month Orders	Order count for the preceding month.
Previous Month Quantity	Quantity sold in the preceding month.

Time-intelligence approach

Previous-month measures use the Date table to shift the date context back one month. For example:

```
Previous Month Sales =
CALCULATE(
    [Total Sales],
    DATEADD('Date'[Date], -1, MONTH)
)
```

The same pattern was applied to orders and quantity. When no single month is selected, a previous-month comparison is not like-for-like, so the Month slicer is set to single selection for those comparisons.

KPI interpretation

- **698.81K Total Sales:** overall revenue for the six-month period.
- **149K Total Orders:** overall order volume.
- **214K Total Quantity:** total items sold.
- **4.69 Average Order Value:** average spend per order.
- **1.44 Average Items per Order:** average basket size. Because order value equals items per order multiplied by average item price, the low basket size is the main driver of the modest order value.

The upselling opportunity is a business hypothesis, not a proven cause. Product-attachment and promotion results would need to be tested before concluding that bundles will increase revenue.

4. Dashboard Overview and Design

The single-page dashboard is organised so a viewer can move from headline numbers to detail: KPIs and title across the top, a Month slicer and three analytical visuals in the second row, and three comparison visuals in the third row.

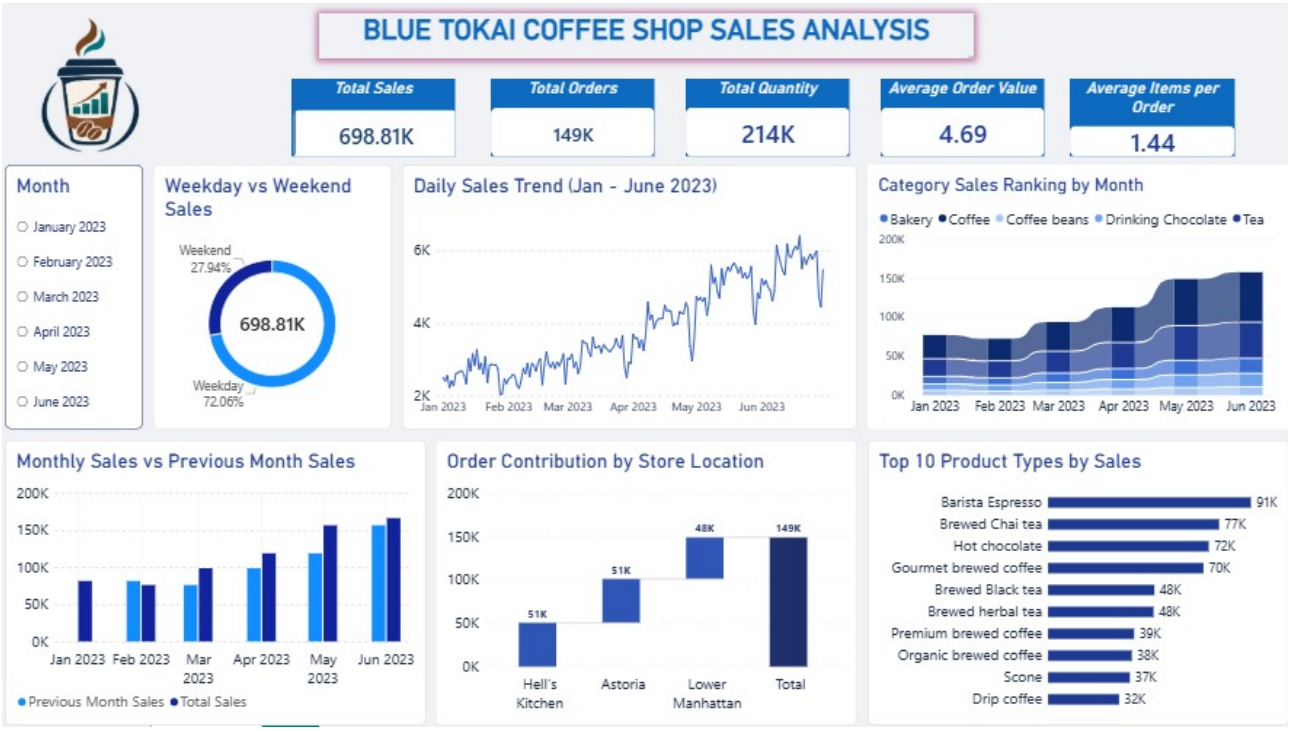


Figure 1. Final Power BI dashboard (January–June 2023, all months selected).

Design decisions

- **Consistent palette:** navy and blue shades on a white background, with navy for current values and lighter blue for comparison values.
- **Aligned grid:** equal visual sizes, equal gaps and shared left and right edges, so the page reads as one block rather than separate charts.
- **Readable text:** a single font family, uniform axis and label colour, and chart titles that state the question each visual answers.
- **Chart type matched to question:** line for trend, ribbon for rank change, clustered column for month comparison, waterfall for contribution to a total, donut for a two-part split, and sorted bar for ranking.
- **Interactivity:** the Month slicer filters the KPI cards and the visuals set to respond to it; the trend visuals keep a full six-month view for context.

5. Dashboard Visuals and Business Questions

The table summarises what each visual asks and what it shows. Detailed interpretation follows.

Visual	Business question	What the viewer learns
KPI cards	How is the business performing overall?	Scale (698.81K sales, 149K orders, 214K items) and customer behaviour (4.69 per order, 1.44 items).
Weekday vs Weekend Sales (donut)	Does revenue come mostly from weekdays or weekends?	Weekdays 72.06% and weekends 27.94% of sales.

Visual	Business question	What the viewer learns
Daily Sales Trend (line)	How are daily sales moving over time?	An upward movement from January to June with daily fluctuations.
Category Sales Ranking by Month (ribbon)	Which categories lead each month, and do rankings change?	Relative contribution and rank of the top five categories across six months.
Monthly Sales vs Previous Month (clustered column)	Is the business growing month over month?	Each month beside the preceding month; overall growth toward June.
Order Contribution by Store (waterfall)	How much does each store contribute to total orders?	About 51K, 51K and 48K orders, building to roughly 149K.
Top 10 Product Types by Sales (bar)	Which products generate the most revenue?	Barista Espresso leads (91K), then Brewed Chai tea (77K), Hot chocolate (72K), Gourmet brewed coffee (70K).

5.1 Weekday vs Weekend Sales — Donut Chart

Business question: Does revenue come mostly from weekdays or weekends?

What it shows: weekdays account for about 72.06% of sales and weekends about 27.94%, with total sales (698.81K) in the centre.

Interpretation: weekdays are five of seven days (about 71.4%), so their share of sales is roughly in line with their share of days. Per trading day, weekdays are marginally stronger (about 14.4 percentage points of sales per weekday against about 14.0 per weekend day). The gap is small, so staffing should be set from exact per-day averages.

5.2 Daily Sales Trend — Line Chart

Business question: How are daily sales moving over time?

What it shows: daily sales rise from roughly the 2K–4K range early in the year to roughly 4K–6K by June, with regular short dips.

Interpretation: sales activity strengthened over the period. The recurring dips are consistent with weekly cycles, but the chart alone does not establish the reason for any individual drop.

5.3 Category Sales Ranking by Month — Ribbon Chart

Business question: Which product categories lead each month, and do their relative positions change?

What it shows: the five highest-selling categories (Bakery, Coffee, Coffee beans, Drinking Chocolate and Tea) across January to June, with ribbons that connect each category between months.

Interpretation: widening ribbons indicate growing contribution, and crossings indicate a change in rank. Exact category winners and growth rates should be read from tooltips before being quoted.

5.4 Monthly Sales vs Previous Month Sales — Clustered Column Chart

Business question: Is the business growing month over month?

What it shows: each month's sales beside the previous month's sales, so the change is visible at a glance.

Interpretation: the overall pattern rises toward June, although February appears slightly below January. The report therefore describes growth across the period, not uninterrupted month-by-month growth. January has no previous-month comparison because the data starts in January 2023.

5.5 Order Contribution by Store Location — Waterfall Chart

Business question: How much does each store contribute to total orders?

What it shows: Hell's Kitchen (about 51K), Astoria (about 51K) and Lower Manhattan (about 48K) build up to a total of about 149K orders, roughly 34%, 34% and 32%.

Interpretation: orders are relatively balanced, so no single location dominates. Lower Manhattan is slightly behind; operating hours, footfall, staffing and local demand would need review to explain the difference.

5.6 Top 10 Product Types by Sales — Horizontal Bar Chart

Business question: Which product types generate the most revenue?

What it shows: Barista Espresso (about 91K), Brewed Chai tea (77K), Hot chocolate (72K) and Gourmet brewed coffee (70K) lead, followed by six types between 32K and 48K.

Interpretation: the top four types together contribute roughly 310K, about 44% of total sales, and the top ten roughly 552K, about 79%. Sales are therefore concentrated in a small set of drinks. These are important for inventory attention and quality consistency, and are natural anchors for bundles, which should be tested before assuming an uplift. (Shares are calculated from rounded dashboard values.)

6. Key Findings and Recommendations

Key findings

- The dashboard reports 698.81K in sales, 149K orders and 214K items for January–June 2023.
- Average order value is 4.69 and average items per order is 1.44.
- Weekdays contribute about 72.06% of sales and weekends about 27.94%, close to their share of calendar days.
- Daily sales trend upward across the six months, despite daily and monthly fluctuations.
- Orders are spread almost evenly across Hell's Kitchen, Astoria and Lower Manhattan.
- Sales are concentrated: the top four product types contribute about 44% of sales and the top ten about 79%.

Business recommendations

- **Test basket-building offers:** pair popular drinks with pastries, scones or other add-ons. Measure attachment rate and average order value before and after the test.
- **Plan staffing by demand per day:** compare average sales and orders per trading day for weekdays and weekends rather than relying on total share.
- **Protect availability of leading products:** review stock and quality for the highest-revenue product types.
- **Investigate store differences:** examine hours, staffing, footfall and product mix at Lower Manhattan.
- **Review monthly changes:** use the Month slicer and previous-month measures to find periods of acceleration or decline, including the February dip.

Overall project summary

Sales increased overall from January to June 2023, reaching 698.81K across about 149K orders. Orders are spread fairly evenly across the three stores, while a small group of product types contributes strongly to revenue. With an average of 1.44 items per order, increasing basket size through relevant add-ons is a practical opportunity to test.

7. Validation, Limitations and Conclusion

Validation checks

- Confirm that the Month slicer filters each intended visual and leaves the trend visuals on the full period.
- Confirm the Date-to-Transactions relationship is active and uses the Date data type on both sides.

- Verify that Previous Month Sales, Orders and Quantity match the prior month's own totals.
- Use tooltips to confirm exact monthly values and percentage changes before quoting them.
- Compare weekday and weekend sales per trading day before drawing staffing conclusions.

Limitations

Figures are taken from the dashboard and are rounded to the nearest K, so derived shares are approximate. The dashboard shows associations and trends; it does not by itself establish why sales changed or prove that a particular action will improve performance. The data covers six months and three stores, so seasonal effects across a full year cannot be assessed.

Possible next steps

- Use **transaction_time** to analyse sales by hour of day and day of week, to support shift planning.
- Analyse which products are bought together to design bundles.
- Extend the date range to compare against the same months in other years.

Conclusion

The project demonstrates Power BI data modelling, DAX measures, time-intelligence calculations, interactive filtering and a set of business-focused visuals. The dashboard gives a clear overview of sales performance and points to practical areas for follow-up: basket size, availability of leading products and store-level demand.